NAME

mbedit-qt − Interactive editor used to flag bad bathymetry values in swath sonar bathymetry data (Qt6 version).

VERSION

Version 5.0

SYNOPSIS

mbedit-qt
[--begin-time=yr/mo/da/hr/mn/sc {-Byr/mo/da/hr/mn/sc}
--browse {-D}
--end-time=yr/mo/da/hr/mn/sc {-Eyr/mo/da/hr/mn/sc}
--format=format_id {-Fformat_id}
--gui-mode {-G}
--help {-H}
--input=file {-Ifile}
--reference=grid
--run-mbprocess {-X}
--use-edit-save {-S}
--verbose {-V}
--dump
--export=image [--size=WxH]
--replay=script]

DESCRIPTION

mbedit-qt is a Qt6 port of **mbedit**. It displays the bathymetry profiles of several pings and lets the user flag and unflag soundings. The edits are written to an edit save file, which **mbprocess** applies when it writes a processed swath file. It reads and writes the same edit save files, parameter files and lock files as **mbedit**, so a file can move between the two programs. The two cannot edit one file at the same time.

The window can be resized, and it follows the light or dark mode of the system.

MB-SYSTEM AUTHORSHIP

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OPTIONS

−−begin−time=yr/mo/da/hr/mn/sc −Byr/mo/da/hr/mn/sc
Pings before this time are ignored. Default: 1962/2/21/10/30/0.

−−browse −D
Starts in browse mode: no edits are written.

−−end−time=yr/mo/da/hr/mn/sc −Eyr/mo/da/hr/mn/sc
Pings after this time are ignored. Default: 2062/2/21/10/30/0.

−−format=format_id −Fformat_id
Sets the **MBIO** format id for the **−−input** options that follow it. By default it is inferred from the file name. A format of −1 means a datalist.

−−gui−mode −G
Done also quits the program, for use when another program starts **mbedit-qt** for one file.

−−help −H
Prints a description of the program and exits.

−−input=file −Ifile
Adds a swath file, or the files of a datalist, to the list of files to edit. The first is loaded at startup. May be given several times.

−−reference=*grid*
Loads this grid as a reference surface (see REFERENCE SURFACE).

−−run−mbprocess −X
Runs **mbprocess** on a file when it is closed, if its edits were saved.

−−use−edit−save −S
Applies an existing edit save file of the first input file without asking.

−−verbose −V
Prints more information to stderr.

−−dump
Without a window, prints one line per ping of the input files: file, record, time, number of beams, and the numbers of good, null, manual, filter, sonar and other flagged beams. Nothing is written.

−−export=image −−size=WxH
Without a window, writes an image of the first pings of the first input file. Nothing else is written. The default size is 1400x900.

−−replay=script
Without a window, runs a script of editing actions, one per line, for testing.

THE WINDOW

Clicking a time series beside the soundings, or the map shown by **Show Track Map** in the **View** menu, moves to that ping. The map draws the navigation of the pings in memory, with those in view marked.

File **Open** adds a swath file or datalist to the list and loads it. **File Selection List** shows the list, with **<loaded>**, **<locked>** and **<esf>** marks; a file can be edited or removed from it. **Browse Only** sets the output mode for the files loaded next. **Save** writes the edits without closing the file.

Edit **Undo** and **Redo** (the usual shortcuts of the system). One click or one drag, or one key or filter run, is one step. Undo reaches back to the last **Next Buffer**, because the pings written then have left memory. Saving does not end it. **Flags by Reason** shows or hides manual, filter and sonar flags, and unflags those of one reason in the pings in view, from the first ping in view, or in all pings in memory. Filter flags of earlier sessions count as filter flags; the edit save file does not record which program made them. Soundings hidden here are not changed by clicks.

View **Waterfall**, **Alongtrack** and **Acrosstrack View**. **Show Flagged Soundings** and **Show Flagged Profile**. Colours by **Flag State** (unflagged in the text colour, manual red, filter blue, sonar green), by **Bottom Detect Algorithm**, by **Source Pulse Type**, by **Amplitude** or by **Beam Angle**, the last two with a colour bar for the soundings in view. Amplitude is offered only if the file has one for each beam; 2% of the values at each end are left out of its scale. **Reset Zoom** sets the plot width from the data again. **Wide Bathymetry Profiles**, **Print Time Stamps**, or a time series beside the soundings: ping interval, longitude, latitude, heading, speed, center beam depth, sonar altitude, sonar depth, roll, pitch

or heave. With the roll, the acrosstrack slope is red and the difference between them blue; if they correlate, there may be a timing error in the roll correction.

Controls

Go To Specified Time finds the first ping later than the time, reading further buffers if needed; if the time is past the end of the file, the file is saved and closed. **Buffer Controls** sets the number of pings held in memory (30000 to start with, limited only by memory; about 40 bytes per sounding) and the number retained at **Next Buffer**. **Annotation** sets the tick intervals. **Filters** opens the filters below. **Reverse Right/Left Key Macros** is for data with beams ordered starboard to port. **Reverse Mouse Buttons** swaps the left and right buttons.

Tools **Next Suspect Ping** and **Previous Suspect Ping** are the review queue, below. **Preview mbclean** runs **mbclean** with the options given on a link to the file in a temporary folder, starting from the current edits, and outlines what it would change; **Accept** takes it as one step of undo, for the pings in memory.

Acrosstrack Width, Vertical Exaggeration, Pings Shown, Pings to Step

Set the plot. The sliders double their range at their upper end. The width is at most 40000 m and the number of pings shown at most 250.

Ping slider

At the bottom of the window. Shows the first ping in view and moves through the pings in memory, one ping per wheel notch. Data are read forward only, so in a file larger than the buffer it cannot go back past the retained pings.

Start, Reverse, Forward, End

Move in the pings in memory, by **Pings to Step**.

Next Buffer

Writes the pings that leave memory and reads more. They cannot be edited again in this session.

Done | Next File

Saves the edits without asking, closes the file, and loads the next in the list if there is one.

Flag View, Unflag View, Unflag Forward

Flag or unflag the soundings in view; the last unflags from the first ping in view to the end of the buffer.

Quit As closing the window.

EDIT MODES

With the left mouse button, and the keys given:

Toggle Q U

The sounding within 50 pixels nearest the click changes state.

Pick W I

The nearest good sounding is flagged.

Erase E O

Good soundings within 15 pixels of the mouse are flagged while the button is down.

Restore R P

As Erase, but flagged soundings are unflagged.

Grab T { [

Good soundings inside the rectangle dragged are flagged.

Lasso H

Good soundings inside the outline drawn are flagged.

Info Y }]
Shows, and prints to stdout, the data of the nearest sounding.

In Toggle, Pick and Info a ring marks the sounding that a click would take.

A ping with good soundings outside the plot has a red label with a small box beside it: clicking the box flags those soundings. A green label means that only flagged soundings are outside. A label is left out where it would overlap the one below; its colour and box remain.

MOUSE AND KEYS

Middle button, right button
One step back, one step forward; more if held for 2 seconds. With **G** held, to the start or the end. **Shift** with the right button opens a menu for the nearest sounding: flag or unflag its ping, center on it, open its water column, and undo or redo.

Wheel
Steps by **Pings to Step**: up goes forward, down goes back.

Command or Control with the wheel, or the pinch of a trackpad
Zooms the width around the mouse.

Shift with the wheel
Changes the vertical exaggeration.

Space with the left button
Moves the plot. **0** puts it back and sets the width from the data.

After zooming the width or moving the plot, what lies outside the plot is hidden: clicks, drags, **Flag View** and **Unflag View** do not change it, and ping labels have no colours or zap boxes. **Bad Ping**, **Good Ping** and the left and right keys still act on the whole ping.

F, V, left and right arrows
One step forward or back; with **G** held, to the end or start. The arrows move one ping.

Z M Flags the good soundings of the last picked ping.

S K Unflags the flagged soundings of the last picked ping.

A J, D L
Flags the good soundings of the last picked ping to the left, or right, of and including the last picked beam.

X < , Flag View.

C > . Unflag View.

! Makes all beams of the last picked ping null. Only **Undo** brings them back. They cannot be unflagged afterwards in **mbedit-qt** or **mbedit**; the data file is not changed.

2 @, 3 #, 4 $
Waterfall, alongtrack and acrosstrack views.

Z, **S**, **A** or **D** held while clicking applies the action to the sounding just picked.

FILTERS

Apply removes the flags of the previous filter runs of this session and runs the filters from the first ping in view to the end of the buffer. Active filters also run on each new buffer. Flags of earlier sessions and of other programs are not removed.

Median Spike
Flags soundings further from the local median depth than a percentage of the sonar altitude. The window has an odd number of beams and of pings.

Wrong Side
Flags soundings on the wrong side of the swath, except within a number of beams from the center.

Flag by Beam Number, Distance, Beam Angle
Flags soundings between a start and an end value, or if the start is above the end, on both edges of the swath.

A **Preview** in the dialog, on by default, outlines what **Apply** would change, orange for new flags and turquoise for flags removed, and gives the numbers. It runs the same steps as **Apply** on a copy of the flags, and nothing changes until then.

REVIEW QUEUE

After each buffer is read, pings are ranked without changing anything: by soundings further than 0.25% of the altitude from the median of nine beams of their ping, by a drop in good soundings against the neighbouring pings, and by jumps in attitude or ping time. The strip above the ping slider shows them, darkest for the worst. **N** goes to the next worst and **B** back; the status bar says why, and the ping label is outlined. Treat this as a suggestion of an order: pings without a mark can still have bad soundings. It is not recalculated after edits, until the next buffer is read.

REFERENCE SURFACE

Load Reference Surface in the **Tools** menu reads a grid in any form that **mbgrid** writes, geographic or projected, or starts with **−−reference**. **Show Difference from Reference** then colours each sounding by its depth minus the depth of the grid, interpolated between the four nearest nodes: red is deeper than the grid, blue is shallower. Green soundings have no grid value. The range of the colours is symmetric about zero and is set by the soundings in view. Whether the grid holds depths or topography (seafloor negative) is decided by comparing it with the soundings, and the status bar says which was found. While a surface is loaded, the review queue also counts soundings further than 2% of the altitude from it.

OTHER PROGRAMS

Edit History lists the versions of the edit save file kept by **mbhistory**. **Preview** outlines what a version would change in the pings in memory. **Restore** closes the file, runs **mbhistory −−restore**, which keeps the current edits as a new version, and reopens the file.

Water Column at Center Ping starts **mbwcdview** on this file or on a Kongsberg file of the same name beside it. The item is there only if **mbwcdview** can be found. It takes no lock, so editing can go on. The ping number is that of this program; **mbwcdview** counts water column pings, so it can show another ping if some pings have no water column data, or if each ping holds two swaths.

Save and Open in mbeditviz closes the file, asking whether to save, and starts **mbeditviz**, which takes the lock. It opens at the start of the file.

SAVING

The existing edit save file is not touched until the edits are saved. Until then new edits go to *file*.esf.new and *file*.esf.new.stream. Saving keeps the old edit save file in the **mbhistory** directory, replaces it and updates the **mbprocess** parameter file.

When the window is closed, the program quits or another file is opened with changes unsaved, the program asks whether to save or discard them, or to cancel. After Discard, the files are as they were before the session. Without changes, nothing is written. If an existing edit save file is not applied when asked, saving replaces it.

After a crash the lock file remains and the edits are in the two files above. When the file is opened again, the program offers to remove the lock and to recover the edits. Beams made null and brought back by undo

before the crash stay null.

DIFFERENCES FROM MBEDIT

Toggle and Restore do not change soundings that are not drawn. An edit save file is written only with changes, and always in explicit mode, even if an earlier one was implicit (**mbgetesf**). Erase and Restore also take the soundings between the mouse positions reported. The roll difference starts from the first ping in view. Ping labels are numbered from 1 in every plot mode. Where the center beam is flagged or null, as is usual in dual swath data, ping labels and the center beam depth plot use the sonar altitude plus the sonar depth instead of its value. The automatic plot width and the width slider stop at 40000 m. **mbprocess** does not block the window. The program name in the edit save file, lock file and history is mbedit-qt.

SEE ALSO

mbsystem(1), **mbedit**(1), **mbprocess**(1), **mbhistory**(1), **mbclean**(1), **mbeditviz**(1), **mbgetesf**(1), **mbhistory**(1), **mbwcdview**(1)

BUGS

Only tested on macOS so far.